

THE RATIOS SYSTEM: ANALYSIS OF ROE (RETURN ON EQUITY)

Chapter 7



Universit 
Bocconi
GRADUATE SCHOOL

- ☐ INTRODUCTION
- ☐ THE DUPONT DECOMPOSITION: THE BASIC FORMULA
- ☐ DUPONT: A BROADER DECOMPOSITION
- ☐ ROE DECOMPOSITION: THE “MODIGLIANI-MILLER” FORMULA
 - “MODIGLIANI-MILLER FORMULA”: FIRST TECHNICAL ADJUSTMENT
 - “MODIGLIANI-MILLER FORMULA”: SECOND TECHNICAL ADJUSTMENT
 - “MODIGLIANI-MILLER FORMULA”: A SUMMARY
- ☐ MORE ON DECOMPOSITION: THE “OPERATING LIABILITY LEVERAGE”
- ☐ EXAMPLE 1 (1/3): THE DATA
 - EXAMPLE 1 - THE “BASIC” ROE FORMULA (I/S AMOUNTS BEFORE TAXES)
 - EXAMPLE 1 - THE “PENMAN” ROE FORMULA (I/S AMOUNTS AFTER TAXES)
- ☐ TWO EXTREME SITUATIONS.....
 - EXAMPLE 2 – NET FINANCIAL ASSETS
 - NET FINANCIAL ASSETS: MICROSOFT ANNUAL REPORT (JUNE 30, 2005)
 - CHANGES IN NET EQUITY AND EFFECTS ON ROE
 - TWO EXTREME SITUATIONS.....1) WHEN NET OPERATING ASSETS ARE NEGATIVE
- ☐ RETURN ON EQUITY AND CREATION OF VALUE
- ☐ EXAMPLE 3 – AN INVESTMENT PROJECT: DCF AND EVA
- ☐ EXAMPLE 4 – A BUSINESS VALUATION: DCF AND EVA
- ☐ EXAMPLE 5 – A FINAL COMPREHENSIVE EXAMPLE



$$\text{R.O.E.} = \frac{\text{Comprehensive Income (C.I.)}}{\text{Net Equity}} = \frac{\text{Net Income}}{\text{Net Equity}} + \frac{\text{(change in) Other Comprehensive Income}}{\text{Net Equity}}$$

- By reformulating the income statement we can eliminate the distinction between net income and O.C.I.
- Decompositions of ROE are based on «additive» formulas to show the components (ratios) that influence the Return on Equity
- There are **several approaches to the ROE decomposition**:
 1. Dupont Formula
 2. Modigliani-Miller Formula, with focus on:
 - ☐ Financial liabilities /Net Equity
 - ☐ Net Financial Obligations /Net Equity
- Moreover, **some formulas can show**:
 - the “**operating liability leverage**” (together with the “financial leverage”)
 - values “**net of (after) taxes**” (instead of values “before taxes”)
 - what to do when Financial Assets are higher than Financial Liabilities (“**Net Financial Assets**”)
 - what to do **when Net Operating Invested Capital is negative?** (i.e. Operating Liabilities are higher than Operating Assets)



$$\text{R.O.E.} = \text{Profit Margin} \times \text{Total Asset Turnover} \times \text{Equity Multiplier}$$

$$\text{R.O.E.} = \frac{\text{Net Income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Total Assets}} \times \frac{\text{Total Assets}}{\text{Common Equity}}$$

This **profitability ratio (profit margin)** give us the information: how much profit a company can generate for each € of sales

This **assets turnover ratio** give us the information: how much money from sales we can generate for each € that is invested in assets of the business

This **financial leverage ratio (equity multiplier)** give us the information: what percentage of total assets employed are covered by shareholder's equity and what percentage by debt

Equity Multiplier	=	1	+	Debt Ratio
$\frac{\text{Total Assets}}{\text{Common Equity}}$	=	1	+	$\frac{\text{Total Liabilities}}{\text{Common Equity}}$

DuPont Corporation started using this formula in the 1920s. DuPont explosives salesman Donaldson Brown invented this formula in an internal efficiency report in 1912.



											4					1					2					3				
ROE	=	NI	X	EBT	X	EBIT	X	Sales	X	Total Assets																				
		EBT		EBIT		Sales		Total Assets		Equity																				
											5																			

1. If EBIT includes both **non operating results** and **unusual items**, its relationship with Net Revenues is quite problematic: the ratio EBIT/Revenues can not be considered a proper measure of “Return on Sales”
2. Moreover, the Assets Turnover ratio can be influenced from **non operating assets** (ex. Financial assets: no correlation with Sales)
3. The “leverage” ratio (assets/equity) is influenced from both **financial liabilities** and **operating liabilities**; again, assets include non operating investments
4. The cost of debt is implicit in the ratio EBT/EBIT; but: where are classified the **financial income**?
5. What happens if we want to separate the **unusual items**?



$$R.O.E. = \frac{\text{Compr. Income}}{\text{Net Equity}} = \left[\frac{\text{Operating Income}}{\text{Invested Capital}} + \left[\frac{\text{Operating Income}}{\text{Invested Capital}} - \frac{\text{Financial Expenses}}{\text{Total Liabilities}} \right] \times \frac{\text{Total Liabilities}}{\text{Net Equity}} \right] \times \frac{\text{Compr. Income}}{\text{Income before Unusual Items and Taxes}}$$

HISTORY:

Franco Modigliani (June 18, 1918 – September 25, 2003) was an Italian economist naturalized American, a professor at the MIT Sloan School of Management and MIT Department of Economics who won the Nobel Memorial Prize in Economics in 1985.

Along with Merton Miller, he formulated the important Modigliani–Miller theorem in corporate finance (1958).

The “famous” Proposition II of the “theorem”:

$$r_E(\text{Levered}) = r_E(\text{Unlevered}) + \frac{D}{E}(r_E(\text{Unlevered}) - r_D)$$

- r_E is the required rate of return on equity, or cost of equity.
- r_D is the required rate of return on borrowings, or cost of debt.
- D/E is the debt-to-equity ratio.

A higher debt-to-equity ratio leads to a higher required return on equity, because of the higher risk involved for equity-holders in a company with debt. The formula is derived from the theory of weighted average cost of capital (WACC).



$$\text{R.O.E.} = \frac{\text{Compr. Income}}{\text{Net Equity}} = \left[\frac{\text{Operating Income}}{\text{Invested Capital}} + \underbrace{\left[\frac{\text{Operating Income}}{\text{Invested Capital}} - \frac{\text{Financial Expenses}}{\text{Total Liabilities}} \right]}_2 \right] \times \frac{\text{Total Liabilities}}{\text{Net Equity}} \times \frac{\text{Compr. Income}}{\text{Income before Unusual Items and Taxes}}$$

1
2
3
4

ROE depends on:

- 1) Operating profitability (return on invested capital),
- 2) Difference (spread) between operating profitability and cost of financial debts
- 3) Debt-to-Equity ratio
- 4) The relevance of unusual items and income taxes ratio

If operating profitability is higher than the cost of debt (2) there is multiplier effect on ROE of the (total)debt-to-equity ratio → for the same spread (ROI - cost of debt), the more a company has debts - the higher the debt-to-equity ratio (3) - the higher is the Return On Equity.....THE FAMOUS FINANCIAL LEVERAGE....

But: the more we have money (borrowing from banks) and.....

- the higher is the financial risk (increase of the cost of debt)
- the lower becomes the operating profitability; assumption: first you invest in the highest return businesses then you invest in lower and lower profitability businesses, i.e. decreasing marginal profitability of operating investments

→ a “big” company can face with a negative spread....DISASTER!

$$\text{R.O.E.} = \frac{\text{Compr. Income}}{\text{Net Equity}} = \left[\frac{\text{Operating Income}}{\text{Invested Capital}} + \underbrace{\left[\frac{\text{Operating Income}}{\text{Invested Capital}} - \frac{\text{Financial Expenses}}{\text{Total Liabilities}} \right]}_2 \times \frac{\text{Total Liabilities}}{\text{Net Equity}} \right] \times \frac{\text{Compr. Income}}{\text{Income before Unusual Items and Taxes}}$$

1
2
3
4

A normal situation:

$$\text{R.O.E.} = 14,4\% = \left[15\% + \underbrace{\left[15\% - 6\% \right]}_2 \times 1 \right] \times 0,6$$

1
2
3
4

A “leveraged” (and profitable) company:

$$\text{R.O.E.} = 25,2\% = \left[15\% + \underbrace{\left[15\% - 6\% \right]}_2 \times 3 \right] \times 0,6$$

1
2
3
4

A “leveraged” (but non-profitable) company:

$$\text{R.O.E.} = -6,0\% = \left[5\% + \underbrace{\left[5\% - 8\% \right]}_2 \times 5 \right] \times 0,6$$

1
2
3
4



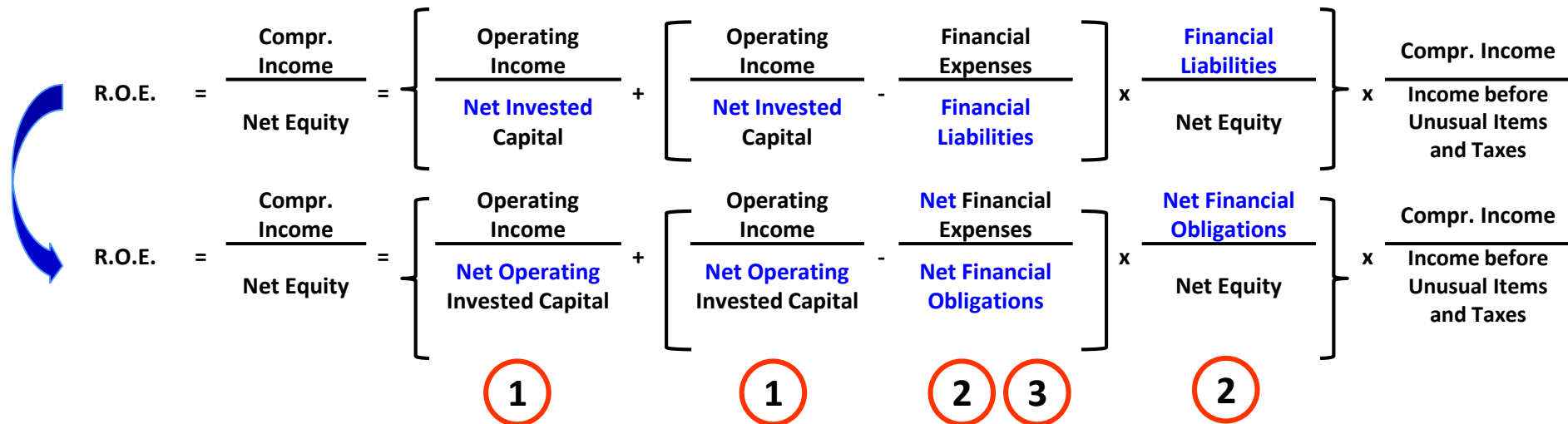
$$\begin{aligned}
 \text{R.O.E.} &= \frac{\text{Compr. Income}}{\text{Net Equity}} = \left[\frac{\text{Operating Income}}{\text{Invested Capital}} + \frac{\text{Operating Income} - \text{Financial Expenses}}{\text{Total Liabilities}} \right] \times \frac{\text{Compr. Income}}{\text{Income before Unusual Items and Taxes}} \\
 \text{R.O.E.} &= \frac{\text{Compr. Income}}{\text{Net Equity}} = \left[\frac{\text{Operating Income}}{\text{Net Invested Capital}} + \frac{\text{Operating Income} - \text{Financial Expenses}}{\text{Financial Liabilities}} \right] \times \frac{\text{Compr. Income}}{\text{Income before Unusual Items and Taxes}}
 \end{aligned}$$

Financial Expenses must be compared with those Liabilities that originate financial interest, i.e. that are explicitly interest-bearing → we must exclude from Total Liabilities the Operating Liabilities, and we get the Financial Liabilities amount.

We need the same adjustment for the Invested Capital. From a financial perspective the operating profitability ratio is based on the amount of Invested Capital that is financed with financial debt and with equity → we deduct the Operating Liabilities from Invested Capital and we get the **Net Invested Capital** amount.

The operating profitability ratio becomes a **Return on Net Assets ratio** (analyst often call this ratio R.O.N.A.). **There are only 3 areas in the reformulated Balance Sheet:**

1. Net Invested Capital (Total Assets – Operating Liabilities)
2. Financial Liabilities
3. Net Equity



The diagram illustrates the reformulation of the Return on Equity (R.O.E.) formula through three steps, indicated by red circles with numbers 1, 2, and 3. A blue curved arrow on the left points from the top formula to the bottom one.

Top Formula (Standard R.O.E.):

$$\text{R.O.E.} = \frac{\text{Compr. Income}}{\text{Net Equity}} = \left[\frac{\text{Operating Income}}{\text{Net Invested Capital}} + \left[\frac{\text{Operating Income}}{\text{Net Invested Capital}} - \frac{\text{Financial Expenses}}{\text{Financial Liabilities}} \right] \times \frac{\text{Financial Liabilities}}{\text{Net Equity}} \right] \times \frac{\text{Compr. Income}}{\text{Income before Unusual Items and Taxes}}$$

Bottom Formula (Reformulated R.O.E.):

$$\text{R.O.E.} = \frac{\text{Compr. Income}}{\text{Net Equity}} = \left[\frac{\text{Operating Income}}{\text{Net Operating Invested Capital}} + \left[\frac{\text{Operating Income}}{\text{Net Operating Invested Capital}} - \frac{\text{Net Financial Expenses}}{\text{Net Financial Obligations}} \right] \times \frac{\text{Net Financial Obligations}}{\text{Net Equity}} \right] \times \frac{\text{Compr. Income}}{\text{Income before Unusual Items and Taxes}}$$

Numbered Steps:

- 1**: Operating Income / Net Operating Invested Capital
- 1**: Operating Income / Net Operating Invested Capital
- 2**: Net Financial Expenses / Net Financial Obligations
- 3**: Net Financial Expenses / Net Financial Obligations
- 2**: Net Financial Obligations / Net Equity

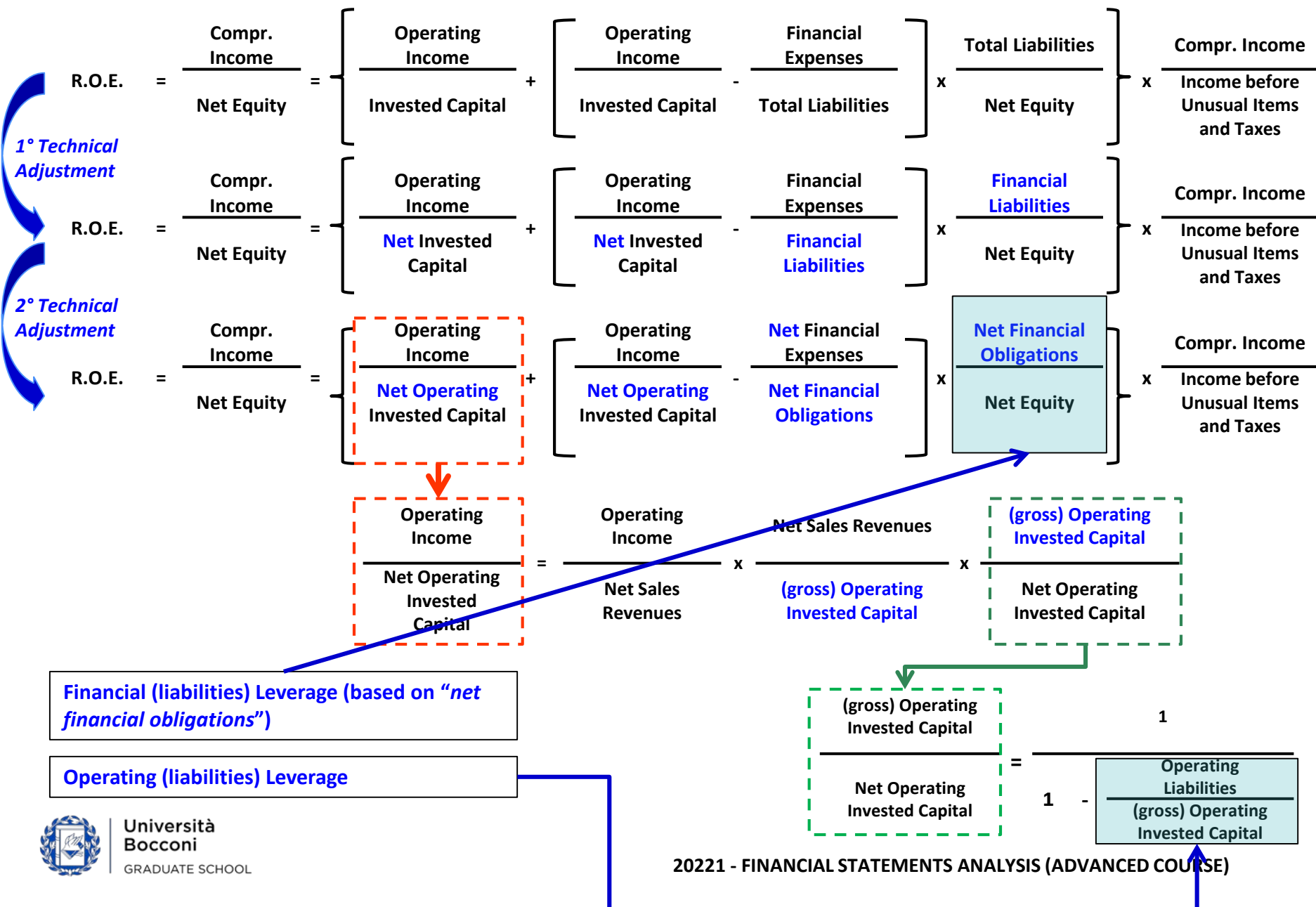
1. If the company has **Financial Investments (and Financial Income)** they should be excluded from the **operating profitability ratio** (RONA) in order to get a ratio that is related to the *stricto sensu* “operating” assets.
2. ➔ We subtract the financial assets from financial liabilities and we get the “**net financial obligations**” amount;
3. ➔ We subtract the financial income from the financial expense and we get the “**net financial expenses**” amount.

Now, the 3 areas of the reformulated B/S are: Net Operating Invested Capital, Net Financial Obligations, Net Equity.

And what about the “**Operating (liabilities) Leverage**”? The ability of the company to borrow money inside the normal operating cycle and finance the operating investments, i.e. substantially:

Operating Liabilities / (gross) Operating Invested Capital





$$\text{R.O.E.} = \frac{\text{Compr. Income}}{\text{Net Equity}} = \left\{ \frac{\text{Operating Income}}{\text{Net Operating Invested Capital}} + \left[\frac{\text{Operating Income}}{\text{Net Operating Invested Capital}} - \frac{\text{Net Financial Expenses}}{\text{Net Financial Obligations}} \right] \times \frac{\text{Net Financial Obligations}}{\text{Net Equity}} \right\} \times \frac{\text{Compr. Income}}{\text{Income before Unusual Items and Taxes}}$$

The diagram illustrates the relationship between Return on Equity (R.O.E.) and Return on Net Operating Assets (RONA), and how they are derived from NOPAT and Net Financial Obligations.

R.O.E. Formula:

$$\text{R.O.E.} = \frac{\text{Compr. Income}}{\text{Net Equity}} = \left[\frac{\text{NOPAT}}{\text{Net Operating Invested Capital}} + \frac{\text{Net Financial Expenses (after taxes)}}{\text{Net Financial Obligations}} \right] \times \frac{\text{Net Financial Obligations}}{\text{Net Equity}} \times \frac{\text{Compr. Income}}{\text{Income before Unusual Items (after taxes)}}$$

NOPAT Formula:

$$\text{NOPAT} = \frac{\text{NOPAT}}{\text{Net Operating Invested Capital}} \times \text{Net Operating Invested Capital}$$

NOPAT Breakdown:

$$\text{NOPAT} = \frac{\text{NOPAT + Implicit Interests (gross)}}{\text{Operating Invested Capital}} \times \text{Operating Invested Capital} - \text{Short-term borrowing rate} \times \text{Operating Liabilities}$$

The diagram uses color coding: Blue for NOPAT, Red for Net Financial Obligations, and Green for Net Equity. Arrows indicate the flow of information and the relationship between the different components.

**Operating
Liabilities x
Short-term
borrowing rate**

The **short-term borrowing cost** can be estimated by calculating the **ratio s-t interest expenses / s-t financial liabilities** (sometimes impossible, if financial statements do not show the distinction of short-term and long-term interest expenses); alternative: use the **market interest rate for short-term borrowings**.

Income Statement:

Net Revenues	1.200,00
Operating Costs	-1.080,00
Operating Income	120,00
Financial Income	4,00
Financial Expenses	-20,00
Income before Unusual Items and Taxes	104,00
Unusual Items	-4,00
Income before Taxes	100,00
Income Taxes (tax rate = 25%)	-25,00
Comprehensive Income	75,00

**Income Statement
(amounts net of taxes):**

NOPAT	90,00
Net Financial Expenses (after taxes)	-12,00
Unusual Items (after taxes)	-3,00
Comprehensive Income	75,00

Balance Sheet:

Operating Invested Capital	800
Financial Assets	200
TOTAL ASSETS	1.000
Operating Liabilities	300
Financial Liabilities	400
Net Equity	300
TOTAL LIABILITIES AD EQUITY	1.000

**Balance Sheet (net amounts):**

Net Operating Invested Capital	500
Net Financial Debts	200
Net Equity	300

Assumption about short-term borrowing rate:

Short-term borrowing rate, after tax	4,00%
Implicit Interest, in Operating Liabilities (300 x 4%)	12



$$\begin{aligned}
 \text{R.O.E.} &= \frac{\text{Net Income}}{\text{Net Equity}} = \left[\frac{\text{Operating Income}}{\text{Net Operating Invested Capital}} + \left(\frac{\text{Operating Income}}{\text{Net Operating Invested Capital}} - \frac{\text{Net Financial Expenses}}{\text{Net Financial Debts}} \right) \times \frac{\text{Net Financial Debts}}{\text{Net Equity}} \right] \times \frac{\text{Income before Extr. Items and Taxes}}{\text{Net Income}} \\
 \text{R.O.E.} &= \frac{75}{300} = \left[\frac{120}{500} + \left(\frac{120}{500} - \frac{16}{200} \right) \times \frac{200}{300} \right] \times \frac{75}{104} \\
 \text{R.O.E.} &= 25,00\% = \left[24,00\% + \left(24,00\% - 8,00\% \right) \times 0,67 \right] \times 0,7212
 \end{aligned}$$

$$\begin{aligned}
 \frac{\text{Operating Income}}{\text{Net Operating Invested Capital}} &= \frac{\text{Operating Income}}{\text{Net Revenues}} \times \frac{\text{Net Revenues}}{\text{Operating Invested Capital}} \times \frac{\text{Operating Invested Capital}}{\text{Net Operating Invested Capital}} \\
 \frac{120}{500} &= \frac{120}{1.200} \times \frac{1.200}{800} \times \frac{800}{500} \\
 24,00\% &= 10,00\% \times 1,50 \times 1,60
 \end{aligned}$$

$$\begin{aligned}
 \frac{\text{Operating Invested Capital}}{\text{Net Operating Invested Capital}} &= \frac{1}{1 - \frac{\text{Operating Liabilities}}{\text{Operating Invested Capital}}} \\
 \frac{800}{500} &= \frac{1}{1 - \frac{300}{800}} \\
 \frac{800}{500} &= \frac{1}{1 - 37,50\%}
 \end{aligned}$$



$$\begin{aligned}
 \text{R.O.E.} &= \frac{\text{Net Income}}{\text{Net Equity}} = \frac{\text{NOPAT}}{\text{Net Operating Invested Capital}} + \left(\frac{\text{NOPAT}}{\text{Net Operating Invested Capital}} - \frac{\text{Net Financial Expenses (after taxes)}}{\text{Net Financial Debts}} \right) \times \frac{\text{Net Financial Debts}}{\text{Net Equity}} \times \frac{\text{Income before Extr.Items (after taxes)}}{\text{Net Income}} \\
 \text{R.O.E.} &= \frac{75}{300} = \frac{90}{500} + \left(\frac{90}{500} - \frac{12}{200} \right) \times \frac{200}{300} \times \frac{75}{78} \\
 \text{R.O.E.} &= 25,00\% = 18,00\% + \left(18,00\% - 6,00\% \right) \times 0,67 \times 0,9615
 \end{aligned}$$

↓

$$\begin{aligned}
 \frac{\text{NOPAT}}{\text{Net Operating Invested Capital}} &= \frac{\text{NOPAT} + \text{Implicit Interests}}{\text{Operating Invested Capital}} + \left(\frac{\text{NOPAT} + \text{Implicit Interests}}{\text{Operating Invested Capital}} - \text{Short-term borrowing rate} \right) \times \frac{\text{Operating Liabilities}}{\text{Net Operating Invested Capital}} \\
 \frac{90}{500} &= \frac{102}{800} + \left(\frac{102}{800} - 4,00\% \right) \times \frac{300}{500} \\
 18,00\% &= 12,75\% + \left(12,75\% - 4,00\% \right) \times 0,60
 \end{aligned}$$





What happens to the formula if:

- Financial assets are higher than financial liabilities? The formula must be modified (see following slides)
- Net Operating Invested Capital is negative? (i.e. Operating Liabilities are higher than Operating Assets)

(A) I/S amounts before taxes

$$\begin{array}{ccccccc}
 \text{Compr. Income} & & \text{Operating Income} & & \text{Operating Income} & \text{Net Financial Revenues} & \text{Net Financial Assets} & & \text{Compr. Income} \\
 & & & & & & & & \\
 \text{R.O.E.} = \frac{\text{Net Equity}}{\text{Net Equity}} = \left[\frac{\text{Operating Income}}{\text{Net Operating Invested Capital}} - \left[\frac{\text{Operating Income}}{\text{Net Operating Invested Capital}} - \frac{\text{Net Financial Revenues}}{\text{Net Financial Assets}} \right] \times \frac{\text{Net Financial Assets}}{\text{Net Equity}} \right] \times \frac{\text{Income before Unusual Items and Taxes}}{\text{Income before Unusual Items and Taxes}}
 \end{array}$$

A positive spread reduces the ROE: return on Financial Assets is usually lower than Return on Operating Assets...

(B) I/S amounts after taxes

$$\begin{array}{ccccccc}
 \text{Compr. Income} & & \text{NOPAT} & & \text{NOPAT} & \text{Net Financial Revenues (after taxes)} & \text{Net Financial Assets} & & \text{Compr. Income} \\
 & & & & & & & & \\
 \text{R.O.E.} = \frac{\text{Net Equity}}{\text{Net Equity}} = \left[\frac{\text{NOPAT}}{\text{Net Operating Invested Capital}} - \left[\frac{\text{NOPAT}}{\text{Net Operating Invested Capital}} - \frac{\text{Net Financial Revenues (after taxes)}}{\text{Net Financial Assets}} \right] \times \frac{\text{Net Financial Assets}}{\text{Net Equity}} \right] \times \frac{\text{Income before Unusual Items (after taxes)}}{\text{Income before Unusual Items (after taxes)}}
 \end{array}$$

Cost of Net Financial Debt is replaced with Return on Net Financial Assets



The same example with a few changes (“red” amounts):

Income Statement:

Net Revenues	1.200,00
Operating Costs	-1.080,00
Operating Income	120,00
Financial Income	4,00
Financial Expenses	-0,00
Income before Unusual Items and Taxes	124,00
Unusual Items	-4,00
Income before Taxes	120,00
Income Taxes (tax rate = 25%)	-30,00
Comprehensive Income	90,00

Income Statement (amounts net of taxes):

NOPAT	90,00
Net Financial Revenues (after taxes)	+3,00
Unusual Items (after taxes)	-3,00
Comprehensive Income	90,00

Balance Sheet:

Operating Invested Capital	800
Financial Assets	200
TOTAL ASSETS	1.000
Operating Liabilities	300
Financial Liabilities	0
Net Equity	700
TOTAL LIABILITIES AD EQUITY	1.000



Balance Sheet (net amounts):

Net Operating Invested Capital	500
Net Financial Assets	200
Net Equity	700

Return on Net Financial Assets is:

- **$4/200 = 2\%$ before taxes**
- **$2\% \times (1-25\%) = 1,5\%$ after taxes**



(A) I/S amounts before taxes

$$\begin{aligned}
 \text{R.O.E.} &= \frac{\text{Compr. Income}}{\text{Net Equity}} = \left[\frac{\text{Operating Income}}{\text{Net Operating Invested Capital}} - \left(\frac{\text{Operating Income}}{\text{Net Operating Invested Capital}} - \frac{\text{Net Financial Revenues}}{\text{Net Financial Assets}} \right) \times \frac{\text{Net Financial Assets}}{\text{Net Equity}} \right] \times \frac{\text{Compr. Income}}{\text{Income before Unusual Items and Taxes}} \\
 \text{R.O.E.} &= \frac{90,00}{700,00} = \left[\frac{120,00}{500,00} - \left(\frac{120,00}{500,00} - \frac{4,00}{200,00} \right) \times \frac{200,00}{700,00} \right] \times \frac{90,00}{124,00} \\
 \text{R.O.E.} &= 12,86\% = \left[24,00\% - \left(24,00\% - 2,00\% \right) \times 0,29 \right] \times 0,7258
 \end{aligned}$$

Identical to the ratio from slide 14

(B) I/S amounts after taxes

$$\begin{aligned}
 \text{R.O.E.} &= \frac{\text{Compr. Income}}{\text{Net Equity}} = \left[\frac{\text{NOPAT}}{\text{Net Operating Invested Capital}} - \left(\frac{\text{NOPAT}}{\text{Net Operating Invested Capital}} - \frac{\text{Net Financial Revenues (after taxes)}}{\text{Net Financial Assets}} \right) \times \frac{\text{Net Financial Assets}}{\text{Net Equity}} \right] \times \frac{\text{Compr. Income}}{\text{Income before Unusual Items (after taxes)}} \\
 \text{R.O.E.} &= \frac{90,00}{700,00} = \frac{90,00}{500,00} - \left(\frac{90,00}{500,00} - \frac{3,00}{200,00} \right) \times \frac{200,00}{700,00} \times \frac{90,00}{93,00} \\
 \text{R.O.E.} &= 12,86\% = 18,00\% - \left(18,00\% - 1,50\% \right) \times 0,29 \times 0,9677
 \end{aligned}$$

Identical to the ratio from slide 15



INCOME STATEMENTS

(In millions, except per share amounts)

Year Ended June 30	2003	2004	2005
Revenue	32.187	36.835	39.788
Operating expenses:			
Cost of revenue	6.059	6.716	6.200
Research and development	6.595	7.779	6.184
Sales and marketing	7.562	8.309	8.677
General and administrative	2.426	4.997	4.166
Total operating expenses	22.642	27.801	25.227
Operating income	9.545	9.034	14.561
Investment income and other	1.509	3.162	2.067
Income before income taxes	11.054	12.196	16.628
Provision for income taxes	3.523	4.028	4.374
Net income	7.531	8.168	12.254

What happened to the Stockholders' Equity in 2005?
Look at the reformulated Balance Sheet:

	2004	2005
Net Operating Assets	2.023	(640)
Net Financial Assets	72.802	48.755
Net Equity	74.825	48.115

STOCKHOLDERS' EQUITY STATEMENTS

(In millions)

Year Ended June 30	2003	2004	2005
Common stock and paid-in capital			
Balance beginning of period	41.845	49.234	56.396
Common stock issued	2.966	2.815	3.223
Common stock repurchased	(691)	(416)	(1.737)
Stock-based compensation expense	3.749	5.734	2.448
Stock option income tax benefits/(deficiencies)	1.365	(989)	89
Other net	-	18	(6)
Balance end of period	49.234	56.396	60.413
Retained earnings (deficit)			
Balance beginning of period	12.997	15.678	18.429
Net income	7.531	8.168	12.254
Other comprehensive income:			
Net gains/(losses) on derivative instruments	(102)	101	(58)
Net unrealized investments gains/(losses)	1.243	(873)	371
Translation adjustments and other	116	51	(6)
Comprehensive income	8.788	7.447	12.561
Common stock cash dividends	(857)	(1.729)	(36.968)
Common stock repurchased	(5.250)	(2.967)	(6.320)
Balance end of period	15.678	18.429	(12.298)
Total stockholders' equity	64.912	74.825	48.115

In addition to dividend distributions, other transactions can also affect the ROE denominator.

Changes of net equity depend on:

- ☐ **Economic performance**

- ☐ **Net Income**

- ☐ **O.C.I.**

- ☐ **Transactions with shareholders:**

- ☐ **Share Issuances:** the issuance of new equity shares by the parent company to external investors increases net equity.

- ☐ **Buybacks:** purchase (sale) of treasury shares decreases (increases) the net equity.

- ☐ **Changes in Ownership Structure (Consolidation Scope & NCI)**

- ☐ **Acquisition of a New Subsidiary:** When a parent acquires control of a company, the net assets are brought into the group. If the parent buys less than 100%, the equity portion belonging to external minority investors is recognized as Non-Controlling Interest (NCI), which increases total consolidated equity.

- ☐ **Changes in Ownership Without Loss of Control:** If a parent buys more shares from or sells shares to the NCI but still maintains legal control, accounting standards require treating this strictly as an equity transaction. No gain or loss is recognized in the P&L; instead, the adjustments are made entirely within the consolidated equity lines.

- ☐ **Loss of Control (Partial or Full Disposal):** Selling a portion of a subsidiary that results in losing control removes that subsidiary's net equity (and its associated NCI) from the consolidated balance sheet.



A CASE OF EXTREME OPERATING LIABILITY LEVERAGE: DELL COMPUTER 2002

Operating assets

Operating cash	25	
Accounts receivable	2.269	
Inventories	278	
Property, plant and equipment	826	
Other assets	1.875	
		5.273

Operating Liabilities

Accounts payable	5.075	
Accrued liabilities	2.444	
Other liabilities	802	
		-8.321

Net Operating Assets -3.048

Net Financial Assets 7.742

Shareholders' Equity 4.694

Net Financial Assets 7,742	Net Operating Assets 3,048
	Net Equity 4,694

Return on Net Operating Assets (RNOA) is negative: Net Operating Assets are - 3.048 !

What happens to this formula?

$$\text{ROE} = \text{RNOA} - \left[\text{NFA} / \text{NE} \times (\text{RNOA} - \text{RNFA}) \right]$$

Even if there are Net Financial Assets, it does not make sense to use a formula where Return on Net Operating Assets is negative. Conceptually, the negative amount of Net Operating Assets, multiplied by the short-term borrowing rate, adds value (and increases the ROE)



- ❑ **Objective:** To demonstrate the relationship between the valuation of an investment and a business using both the DCF (Discounted Cash Flow) method and the EVA (Economic Value Added) method.
- ❑ **The EVA method is based on the concept of residual income, or the operating income remaining after the cost of capital has been remunerated.**
- ❑ **Conceptually, the two methods yield the same results when the changes in invested capital (in the EVA method) equal the cash flows from capex and changes in net operating working capital (in the DCF method).**
- ❑ **We will see below that the breakdown of ROE (accounting return on equity) can also provide insights into the concept of residual income and the creation of economic value (as in the EVA method) when comparing the values obtained with the cost of equity.**



Consider a 500 investment (CAPEX), 4 years horizon, straight line depreciation, 10% WACC:
See below for the annual NOPAT and change in working capital ("red" numbers)

DCF (Discounted Cash Flow):

DCF (Discounted Cash Flow):	Year --> 0	1	2	3	4	
EBIT (1 - tax rate) = NOPAT		75,0	75,0	75,0	75,0	
+ Depreciation		125,0	125,0	125,0	125,0	
-/+ Increase/Decrease in NOWC		-20,0	-5,0	15,0	10,0	
Operating Cash Flow, after taxes		180,0	195,0	215,0	210,0	
- Investment, net (CAPEX)	-500,0	0,0	0,0	0,0	0,0	
Operating Free Cash Flow	-500,0	180,0	195,0	215,0	210,0	
Discount Factor, at a wacc of:	10%	1,0000	0,9091	0,8264	0,7513	0,6830
		-500,0	163,6	161,2	161,5	143,4

NPV at 10,0%

\$129,76

of all the projected cash flows
(including the 500 investment at time 0)

EVA (Economic Value Added):

	Year --> 0	1	2	3	4
Beginning Invested Capital	0,0	500,0	395,0	275,0	135,0
+ CAPEX	500,0	0,0	0,0	0,0	0,0
- Depreciation		-125,0	-125,0	-125,0	-125,0
+/- Increase/Decrease in NOWC		20,0	5,0	-15,0	-10,0
Ending Invested Capital	500,0	395,0	275,0	135,0	0,0
EBIT (1 - tax rate) = NOPAT		75,0	75,0	75,0	75,0
WACC x Capital		-50,0	-39,5	-27,5	-13,5
EVA (Economic Value Added)		25,0	35,5	47,5	61,5
Discount Factor, at a wacc of:	10%	0,9091	0,8264	0,7513	0,6830
		22,7	29,3	35,7	42,0

NPV at 10,0%

\$129,76

of the projected EVAs
(excluding the 500 investment at time 0)



Assumptions:

Assumptions:

		0	1	2	TIME 3	4	5	6 + (TV)	
Revenues		790,0	800,0	810,0	815,0	820,0	825,0	825,0	
Invested Capital - beginning			1.000,0	1.035,0	1.071,1	1.105,7	1.141,4	1.178,1	
- Depreciation (% on beg. IC)	5%		-50,0	-51,8	-53,6	-55,3	-57,1	-58,9	No increase (decrease) in NOWC
+/- Increase/Decrease in NOWC (% on Revenues change)	50%		5,0	5,0	2,5	2,5	2,5	0,0	
+ CAPEX (% on beg. IC)	8%		80,0	82,8	85,7	88,5	91,3	58,9	No growth in Invested Capital : CAPEX = Depreciation
Invested Capital - ending		1.000,0	1.035,0	1.071,1	1.105,7	1.141,4	1.178,1	1.178,1	
Return on (beginning) Invested Capital		15,00%	15,00%	15,00%	15,00%	15,00%	15,00%	15,00%	

No increase (decrease) in NOWC

No growth in Invested Capital :
CAPEX = Depreciation

DCF evaluation:

		TIME						
		0	1	2	3	4	5	6 + (TV)
DCF evaluation:	Operating Income		150,0	155,3	160,7	165,9	171,2	176,7
	Tax Rate	35%	-52,5	-54,3	-56,2	-58,0	-59,9	-61,8
	NOPAT		97,5	100,9	104,4	107,8	111,3	114,9
	+ Depreciation		50,0	51,8	53,6	55,3	57,1	58,9
	-/+ Increase/Decrease in NOWC		-5,0	-5,0	-2,5	-2,5	-2,5	0,0
	Operating Cash Flow		142,5	147,7	155,5	160,6	165,8	173,8
	- CAPEX (Net Investments in Tangible and Intangible Assets)		-80,0	-82,8	-85,7	-88,5	-91,3	-58,9
	Free Operating Cash Flow		62,5	64,9	69,8	72,1	74,5	114,9
	Discount Factor - WACC at:	8%	0,9259	0,8573	0,7938	0,7350	0,6806	0,6806
	NPV of FOCFs		57,9	55,6	55,4	53,0	50,7	
	Terminal Value (No Growth rate)							1.435,8
	NPV of Terminal Value							977,2
Enterprise Value: DCF Evaluation			57,9	55,6	55,4	53,0	50,7	977,2
								1.249,8

No growth (decline) in NOWC

No growth in Invested Capital
CAPEX = Depreciation

FOCF / wacc
114,9 / 8 %
(perpetuity without a growth rate)

TV x Discount Factor
(year 6th)
1.435,6 x 0,6806

No growth (decline) in NOWC

No growth in Invested Capital
CAPEX = Depreciation

FOCF / wacc
114,9 / 8 %
(perpetuity without a growth rate)

TV x Discount Factor
(year 6th)
1.435,6 x 0,6806



Assumptions:

Assumptions:		TIME						
		0	1	2	3	4	5	6 + (TV)
Revenues		790,0	800,0	810,0	815,0	820,0	825,0	825,0
Invested Capital - beginning			1.000,0	1.035,0	1.071,1	1.105,7	1.141,4	1.178,1
- Depreciation (% on beg. IC)	5%		-50,0	-51,8	-53,6	-55,3	-57,1	-58,9
+/- Increase/Decrease in NOWC (% on Revenues change)	50%		5,0	5,0	2,5	2,5	2,5	0,0
+ CAPEX (% on beg. IC)	8%		80,0	82,8	85,7	88,5	91,3	58,9
Invested Capital - ending		1.000,0	1.035,0	1.071,1	1.105,7	1.141,4	1.178,1	1.178,1
Return on (beginning) Invested Capital		15,00%	15,00%	15,00%	15,00%	15,00%	15,00%	15,00%

No increase (decrease) in NOWC

No growth in Invested Capital : CAPEX = Depreciation

EVA evaluation:

EVA evaluation:		TIME						
		0	1	2	3	4	5	6 +
NOPAT			97,5	100,9	104,4	107,8	111,3	114,9
Invested Capital - beginning			1.000,0	1.035,0	1.071,1	1.105,7	1.141,4	1.178,1
Wacc x Capital Employed)	8%		-80,0	-82,8	-85,7	-88,5	-91,3	-94,2
EVA			17,5	18,1	18,7	19,3	20,0	20,6
Discount Factor - WACC at:	8%		0,9259	0,8573	0,7938	0,7350	0,6806	0,6806
NPV of EVA			16,2	15,5	14,9	14,2	13,6	
Terminal Value (No Growth rate)								257,7
NPV of Terminal Value								175,4
Enterprise Value: EVA Evaluation		1.000,0	16,2	15,5	14,9	14,2	13,6	175,4

EVA / wacc
20,6 / 8 %
(perpetuity without a growth rate)



Income Statement

Sales Revenues	1.800,00
Operating Income	180,00
Financial Income	24,00
Financial Expenses	-60,00
Income before Unusual Items and Taxes	
Taxes	144,00
Unusual Items	50,00
Income before Taxes	194,00
Income Taxes (tax rate 30%)	-58,20
Comprehensive Income	135,80



NOPAT	126,00
Net Financial Expenses (after taxes)	-25,20
Unusual Items (after taxes)	35,00
Comprehensive Income	135,80

Balance Sheet

Operating Invested Capital	1.200
Financial Assets	800
TOTAL ASSETS	2.000
Operating Liabilities	600
Financial Liabilities	1.000
Net Equity	400
TOTAL LIABILITIES AD EQUITY	2.000



Net Operating Invested Capital	600
Net Financial Debts	-200
Net Equity	400



Sales Revenues	1.800,00
Operating Income	180,00
Financial Income	24,00
Financial Expenses	-60,00
Income before Unusual Items and Taxes	
Taxes	144,00
Unusual Items	50,00
Income before Taxes	194,00
Income Taxes (tax rate 30%)	-58,20
Comprehensive Income	135,80

Operating Invested Capital	1.200
Financial Assets	800
TOTAL ASSETS	2.000
Operating Liabilities	600
Financial Liabilities	1.000
Net Equity	400
TOTAL LIABILITIES AD EQUITY	2.000

Net Operating Invested Capital	600
Net Financial Debts	-200
Net Equity	400

The Data

$$\begin{aligned}
 \text{R.O.E.} &= \frac{\text{Comprehens. Income}}{\text{Net Equity}} = \left[\frac{\text{Operating Income}}{\text{Net Operating Invested Capital}} + \left(\frac{\text{Operating Income}}{\text{Net Operating Invested Capital}} - \frac{\text{Net Financial Expenses}}{\text{Net Financial Debts}} \right) \times \frac{\text{Net Financial Debts}}{\text{Net Equity}} \right] \times \frac{\text{Comprehens. Income}}{\text{Income before Unusual Items and Taxes}} \\
 &= \frac{135,80}{400,00} = \left[\frac{180,00}{600,00} + \left(\frac{180,00}{600,00} - \frac{36,00}{200,00} \right) \times \frac{200,00}{400,00} \right] \times \frac{135,80}{144,00} \\
 &= 33,95\% = [30,00\% + (30,00\% - 18,00\%) \times 0,50] \times 0,94
 \end{aligned}$$

Even if the two components make sense

- Financial Income / Financial Assets = 24/800 = 3%
- Financial Expenses / Financial Liabilities = 60/1.000=6%
- ... when we offset them we get a crazy number! (18%)

I don't like this number

This spread doesn't make sense!



Università
Bocconi

GRADUATE SCHOOL

Sales Revenues	1.800,00
Operating Income	180,00
Financial Income	24,00
Financial Expenses	-60,00
Income before Unusual Items and Taxes	
Taxes	144,00
Unusual Items	50,00
Income before Taxes	194,00
Income Taxes (tax rate 30%)	-58,20
Comprehensive Income	135,80

Operating Invested Capital	1.200
Financial Assets	800
TOTAL ASSETS	2.000
Operating Liabilities	600
Financial Liabilities	1.000
Net Equity	400
TOTAL LIABILITIES AD EQUITY	2.000

Net Operating Invested Capital	600
Net Financial Debts	-200
Net Equity	400

The Data

$$\begin{aligned}
 \text{R.O.E.} &= \frac{\text{Comprehens. Income}}{\text{Net Equity}} = \left[\frac{\text{Operating Income (+ Financial Income)}}{\text{Net Invested Capital (+ Financial Assets)}} + \left(\frac{\text{Operating Income (+ Financial Income)}}{\text{Net Invested Capital (+ Financial Assets)}} - \frac{\text{Financial Expenses}}{\text{Financial Debt}} \right) \times \frac{\text{Financial Debt}}{\text{Net Equity}} \right] \times \frac{\text{Comprehens. Income}}{\text{Income before Unusual Items and Taxes}} \\
 &= \frac{135,80}{400,00} = \left[\frac{204,00}{1.400,00} + \left(\frac{204,00}{1.400,00} - \frac{60,00}{1.000,00} \right) \times \frac{1.000,00}{400,00} \right] \times \frac{135,80}{144,00} \\
 &= 33,95\% = [14,57\% + (14,57\% - 6,00\%) \times 2,50] \times 0,94
 \end{aligned}$$

The profitability of assets depends now on both operating assets (1.200) and financial assets (200): but we know how to solve this problem!

The cost of (gross) Financial Debts is now a good number!



$$\begin{aligned}
 \text{R.O.E.} &= \frac{\text{Comprehens. Income}}{\text{Net Equity}} = \left[\frac{\text{Operating Income (+ Financial Income)}}{\text{Net Invested Capital (+ Financial Assets)}} + \left(\frac{\text{Operating Income (+ Financial Income)}}{\text{Net Invested Capital (+ Financial Assets)}} - \frac{\text{Financial Expenses}}{\text{Financial Debt}} \right) \times \frac{\text{Financial Debt}}{\text{Net Equity}} \right] \times \frac{\text{Income before Unusual Items and Taxes}}{\text{Comprehens. Income}} \\
 &= \frac{135,80}{400,00} = \left[\frac{204,00}{1.400,00} + \left(\frac{204,00}{1.400,00} - \frac{60,00}{1.000,00} \right) \times \frac{1.000,00}{400,00} \right] \times \frac{135,80}{144,00} \\
 &= 33,95\% = [14,57\% + (14,57\% - 6,00\%) \times 2,50] \times 0,94
 \end{aligned}$$

This number is disturbing me....

Another offsetting effect...

- Unusual Items = +50,0

- Income Taxes = -58,2

Let's split them!

		Core Operating Profitability			Non Core Operating Profitability	
		Operating Income	Net Operating Invested Capital		Financial Income	Financial Assets
Operating Income (+ Financial Income)	=	Operating Income	x	Net Operating Invested Capital	+	Financial Income
Net Invested Capital (+ Financial Assets)		Net Operating Invested Capital		Net Invested Capital (+ Financial Assets)		Net Invested Capital (+ Financial Assets)
$\frac{204,00}{1.400,00}$	=	$\frac{180,00}{600,00}$	x	$\frac{600,00}{1.400,00}$	x	$\frac{24,00}{800,00}$
14,57%	=	30,00%	x	42,86%	x	3,00%



$$\begin{aligned}
 \text{R.O.E.} &= \frac{\text{Comprehens. Income}}{\text{Net Equity}} = \left[\frac{\text{Operating Income (+ Financial Income)}}{\text{Net Invested Capital (+ Financial Assets)}} + \left(\frac{\text{Operating Income (+ Financial Income)}}{\text{Net Invested Capital (+ Financial Assets)}} - \frac{\text{Financial Expenses}}{\text{Financial Debt}} \right) \times \frac{\text{Financial Debt}}{\text{Net Equity}} \right] \times \frac{\text{Comprehens. Income}}{\text{Income before Unusual Items and Taxes}} \\
 &= \frac{135,80}{400,00} = \left[\frac{204,00}{1.400,00} + \left(\frac{204,00}{1.400,00} - \frac{60,00}{1.000,00} \right) \times \frac{1.000,00}{400,00} \right] \times \frac{135,80}{144,00} \\
 &= 33,95\% = [14,57\% + (14,57\% - 6,00\%) \times 2,50] \times 0,94
 \end{aligned}$$

$$\begin{aligned}
 \frac{\text{Comprehens. Income}}{\text{Income before Unusual Items and Taxes}} &= \frac{\text{Comprehens. Income}}{\text{Earnings before Taxes}} \times \frac{\text{Earnings before Taxes}}{\text{Income before Unusual Items and Taxes}} \\
 \frac{135,80}{144,00} &= \frac{135,80}{194,00} \times \frac{194,00}{144,00} \\
 0,94 &= 0,70 \times 1,35
 \end{aligned}$$

Remember...average tax rate was 30%



Now we know
the way

$$R.O.E. = \frac{\text{Comprehens. Income}}{\text{Net Equity}} = \left[\frac{\text{Operating Income (+ Financial Income)}}{\text{Net Invested Capital (+ Financial Assets)}} + \left(\frac{\text{Operating Income (+ Financial Income)}}{\text{Net Invested Capital (+ Financial Assets)}} - \frac{\text{Financial Expenses}}{\text{Financial Debt}} \right) \times \frac{\text{Financial Debt}}{\text{Net Equity}} \right] \times \frac{\text{Comprehens. Income}}{\text{Income before Unusual Items and Taxes}}$$

$$= \frac{135,80}{400,00} = \left[\frac{204,00}{1.400,00} + \left(\frac{204,00}{1.400,00} - \frac{60,00}{1.000,00} \right) \times \frac{1.000,00}{400,00} \right] \times \frac{135,80}{144,00}$$

$$= 33,95\% = [14,57\% + (14,57\% - 6,00\%) \times 2,50] \times 0,94$$

Operating Income (+ Financial Income)		Operating Income		Net Operating Invested Capital		Financial Income		Financial Assets
Net Invested Capital (+ Financial Assets)	=	Net Operating Invested Capital	x	Net Invested Capital (+ Financial Assets)	+	Financial Assets	x	Net Invested Capital (+ Financial Assets)
204,00	=	180,00	x	600,00	x	24,00	x	800,00
1.400,00	=	600,00	x	1.400,00	x	800,00	x	1.400,00
14,57%	=	30,00%	x	42,86%	x	3,00%	x	57,14%

Comprehens. Income		Comprehens. Income		Earnings before Taxes
Income before Unusual Items and Taxes	=	Earnings before Taxes	x	Income before Unusual Items and Taxes
135,80	=	135,80	x	194,00
144,00	=	194,00	x	144,00
0,94	=	0,70	x	1,35

Operating Income		Operating Income		Sales Revenues		Gross Operating Invested Capital
Net Operating Invested Capital	=	Sales Revenues	x	Gross Operating Invested Capital	x	Net Operating Invested Capital
180,00	=	180,00	x	1.800,00	x	1.200,00
600,00	=	1.800,00	x	1.200,00	x	600,00
30,00%	=	10,00%	x	1,50	x	2,00

- They use the financial leverage (spread = 8,57% and leverage = 2,50)
- Return on Net IC (14,57%) depends on a good Core Operating Profitability (30%) but is depressed by a low return on financial assets (3%) where the company invests 57,14% of the total net assets in financial instruments
- But the Core Operating Profitability seems good (30%): in fact the RoS is only 10%; there is an "operating" benefit from the Assets Turnover (1,50) and a further "financial" benefit from the relevance of operating liabilities (they finance 50% of total operating assets): the return on gross operating assets is therefore 15% (10,00% x 1,50)
- Average tax cost is 30% and Unusual Items increase the EBT by 35%

$$\begin{aligned}
 \text{R.O.E.} &= \frac{\text{Comprehens. Income}}{\text{Net Equity}} = \left[\frac{\text{Operating Income (+ Financial Income)}}{\text{Net Invested Capital (+ Financial Assets)}} + \left(\frac{\text{Operating Income (+ Financial Income)}}{\text{Net Invested Capital (+ Financial Assets)}} - \frac{\text{Financial Expenses}}{\text{Financial Debt}} \right) \times \frac{\text{Financial Debt}}{\text{Net Equity}} \right] \times \frac{\text{Comprehens. Income}}{\text{Income before Unusual Items and Taxes}} \\
 &= \frac{135,80}{400,00} = \left[\frac{204,00}{1.400,00} + \left(\frac{204,00}{1.400,00} - \frac{60,00}{1.000,00} \right) \times \frac{1.000,00}{400,00} \right] \times \frac{135,80}{144,00} \\
 &= 33,95\% = [14,57\% + (14,57\% - 6,00\%) \times 2,50] \times 0,94
 \end{aligned}$$

ROE before Unusual Items and Taxes = 36,00%

IS THE COMPANY ADDING VALUE? To answer this question we need to know what is the expected return from the perspective of shareholders...the Wacc! Assuming a Cost of Equity (before taxes) of 20% (we already know that the Cost of Debt is 6%), Wacc (before taxes) will be 10%:

K_d	6,0%	→	1.000	71,4%
K_e	20,0%	→	400	28,6%
wacc	10,0%	→	1.400	100,0%

$$\begin{aligned}
 &\text{ROIC (or ROCE)} > \text{Wacc} \\
 &14,57\% > 10,0\%
 \end{aligned}$$

That's the concept of Residual Income

$$\begin{aligned}
 &\text{Net Equity} \times (\text{ROE}_{\text{BTUI}} - K_e) = \text{EVA} \\
 &400 \times (36\% - 20\%) = 64
 \end{aligned}$$

$$\begin{aligned}
 &\text{Net Invested Capital} \times (\text{ROIC} - \text{Wacc}) = \text{EVA} \\
 &1.400 \times 4,57\% = 64
 \end{aligned}$$

